COM 106 Electrical Engineering

Associate Professor Ir. Ts. Dr. Wong Shen Yuong

Xiamen University Malaysia

Objectives

This course aims to equip students with the abilities:

- To learn the basic concepts of electrical systems in modern world.
- To recognize the interrelationship between electrical engineering and other fields of science and engineering
- To understand fundamental principles of voltage, current, power in the design and analysis of electrical system
- To develop various skills required for the electrical power engineering analysis.

Course Learning Outcomes

Upon the completion of the course, the students are able to:

CLO 1- Demonstrate the understanding of the circuit elements in the electrical system.

CLO 2- Apply various circuit analysis for solving electrical engineering problems.

CLO 3- Solve scientifically and critically for practical problems of electrical engineering.

Programme Learning Outcome

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3					√							

Programme Learning Outcome

PLO1	Engineering Knowledge - Apply knowledge of mathematics, science, electrical & electronic engineering fundamentals and engineering specialisation principles to define and utilize engineering procedures, processes, systems and methodologies.
PLO2	Problem Analysis - Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PLO3	Design/Development of Solutions - Design solutions for complex engineering problems and systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO4	Investigation - Conduct investigation into complex problems using research based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PLO5	Modern Tool Usage - Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.
PLO6	The Engineer and Society - Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems (original PLO8)

Program Educational Objectives

1. **Professionalism**-Graduates will be capable of using problem-solving techniques and critical analysis to establish themselves as practicing professionals and demonstrate leadership skills and entrepreneurial spirit in the field of Electrical and Electronics Engineering or related fields.
2. **Continuous Personal Development**-Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for industrial and academic careers.
3. **Societal Engagement**-Graduates will contribute to sustainable development and the well-being of society.

RULES

- Attendance is **COMPULSORY**
- Absent – medical certificates or related documents are mandatory
- Submit all assignments

Content Outline

	Content Outline
1	Introduction to the Course
2	Introduction to Electrical Engineering
3	Introduction to Circuit Elements
4	Electrical Laws
5	Resistive Circuits
6	Electrical Circuit Analysis
7	Capacitance and Inductance
8	Steady State Sinusoidal Analysis

Course in detail:

Topic 1:

Overview of Electrical Engineering

- Concept of circuit, voltage, current
- Concept of power and energy

Topic2:

Introduction to circuit element

- Concept of independent and dependent sources
- Ohm’s law
- Kirchhoff’s laws

Course Outline: (cont.)

Topic 3:
Resistive circuit

- Resistors in series and parallel
- Voltage divider and current divider

Topic 4:
Methods of analysis in circuit

- Nodal analysis
- Application of supernode in nodal analysis

Topic 5:

- Mesh analysis
- Application of supermesh in mesh analysis

Course Outline: (cont.)

Topic 6:

Useful circuit theorems

- Thevenin's theorem
- Norton's theorem
- Superposition theorem

Topic 7:

- Source Transformation
- Max power transfer
- Superposition

Course Outline: (cont.)

Topic 8:

- Capacitance
- Capacitance in series and parallel
- Inductance
- Inductance in series and parallel

Topic 9:

Introduction to sinusoidal circuit

- Sinusoidal current and voltages
- Phasors
- Complex impedances
- Circuit Analysis with phasors and complex impedances

Assessment

Method	%
Assignment	30
Quiz/Test	20
Final Exam	50
TOTAL	**100**

Grading System

Grade	Marks	Grade Point	Status
A	85 – 100	4.0	Distinction
A-	80 – 84	3.7	Distinction
B+	75 – 79	3.3	Good
B	70 – 74	3.0	Good
B-	65 – 69	2.7	Good
C+	60 – 64	2.3	Pass
C	55 – 59	2.0	Pass
F	Below 55	0	Fail

Note: you required to pass in the Final Examination to get overall pass!

Attendance

- You are required to attend the class for more than **80%** of the total credit hours

- If not, you will be **barred** from the final examination/ affect your coursework marks

Contacts

- Email : shenyuong.wong@xmu.edu.my
- Room : A1-436
- Phone : 03-87055068

References

- Allan R Hambley, Electrical Engineering: Principles and Applications, International Edition, Pearson, 2018

